An Offset Cancellation Technique for Two-Stage CMOS Operational Amplifiers

Chih-Wen Lu

***Abstract*—An offset cancellation technique for two-stage CMOS operational amplifiers is proposed. The auxiliary amplifiers are employed to cancel the offset voltage. The error voltage is stored on a capacitor during the cancellation phase and is canceled during the normal operation. An experimental prototype amplifier implemented in a 0.35-μm CMOS technology demonstrates that the offset voltage is reduced to 1.898 mV under a clock frequency of 200 KHz.**

***Index Terms*—offset, offset voltage, offset cancellation, operational amplifier, op-amp.**

I. INTRODUCTION

THE CMOS operational amplifier is the most important building block of linear CMOS and switched-capacitor circuits [1-3, 7]. However, mismatch of the devices causes an offset voltage, which limits the high-precision application [4-6]. Some methods were proposed to reduce the offset voltage. For example, "output offset storage" and "input offset storage" techniques measure the offset voltage and store the result on capacitors in series with the output/input [7]. The general drawback of these two techniques is that they introduce capacitors in the signal path, a particularly serious issue in op-amps and feedback system [7].

To resolve the above issues, an offset cancellation scheme, which can isolate the signal path from the offset storage capacitors through the use of an auxiliary amplifier, was proposed. Fig. 1 shows this conventional architecture of offset cancellation where each G_m stage is a differential pair and the R stage represents a transimpedance amplifier [7]. G_{m1} and R may constitute a one-stage op-amp, which is the main amplifier. G_{m2} is an added auxiliary amplifier, which is used to store the offset voltage. V_{OS1} and V_{OS2} are the input-referred offset voltages for the main and auxiliary amplifiers, respectively. After the offset cancellation, these two offset voltages will be reduced. The total reduced offset voltage, $V_{OS,tot}$, can be expressed as:

$$V_{OS,tot} = \frac{V_{OS1}}{G_{m2}R} + \frac{V_{OS2}}{G_{m1}R}. \quad (1)$$

If $G_{m2}R$ and $G_{m1}R$ are large, then the total offset voltage is very small. Upon turning off, S_3 and S_4 may inject unequal charges onto C_1 and C_2, respectively, creating an error voltage that is not corrected. The input-referred offset voltage, which is due to the injection error, is expressed:

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$$V_{OS,inj} = \left(\frac{G_{m2}}{G_{m1}}\right)\Delta V \quad (2)$$

where $\triangle V$ represents a differential injection-induced error voltage. To reduce the injection error, the value of G_{m2} can not be chosen to large. However, $V_{OS,tot}$ will be larger for a smaller value of G_{m2}. Furthermore, the smaller devices of the differential pair of the auxiliary amplifier usually cause a larger value of the input referred offset voltage, i.e. V_{OS2}. Hence, this is a trade off between the offset voltage and the injection error. In order to increase the gain for this one-stage amplifier, the transimpedance stage should be implemented by a cascode configuration. This limits the minimum value of the power supply. As the power supply voltage is continued to scale down, fewer cascode configuration is used for low voltage supply amplifier and more cascade circuits are employed in two-stage op-amps. In this work, an offset cancellation technique, which can be used in low-voltage two-stage CMOS op-amps, is proposed.

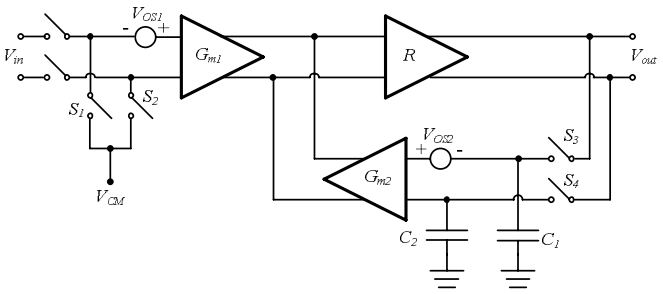


Fig. 1 The conventional architecture of offset cancellation.

II. PROPOSED OFFSET CANCELLATION TECHNIQUE

A. Architecture

Fig. 2 shows the proposed offset cancellation architecture with a feedback loop consisting of a transimpedance, R_1, an amplifier, A_2, and auxiliary amplifiers, A_{aux} and $G_{m,aux}$. G_{m1}, R_1, and A_2 form the main two-stage op-amp. G_{m1} is a differential pair. R_1 is the active load of G_{m1}. Offset voltages, V_{OS1} and V_{OS2} are the input-referred offset voltages for the main and auxiliary amplifiers, respectively. During the offset cancellation, the switches, S_1 and S_2, are turned on. A negative feedback loop is formed. The input voltage is applied to the non-inverting terminals of G_{m1} and A_{aux}. As depicted in Fig. 2, the relation between the input and output voltages can be expressed as:

$$\left[\pm V_{OS1}G_{m1} - \left(V_{in}^{+} \pm V_{OS2} - V_{out}\right)A_{aux}G_{m,aux}\right]R_1A_2 = V_{out}. \quad (3)$$

Thus,

$$\begin{aligned} V_{out} &= \frac{\left(V_{in}^{+} + V_{OS2}\right)A_{auz}G_{m,aux}R_1A_2 \pm V_{OS1}G_{m1}R_1A_2}{A_{aux}G_{m,aux}R_1A_2 - 1} \\ &\cong \pm V_{in}^{+} + V_{OS2} \pm \frac{V_{OS1}G_{m1}}{A_{aux}G_{m,aux}} \end{aligned} \quad (4)$$

Then the output offset voltage is that

$$V_{out} - V_{in}^{+} = \pm V_{OS2} \pm \frac{V_{OS1}G_{m1}}{A_{aux}G_{m,aux}}. \quad (5)$$

A voltage of A_{aux} V_{out} is stored on the storage capacitor C after the switches are turned off. The offset voltage referred to the main amplifier is therefore given by

$$\begin{aligned} V_{OS,tot} &= \frac{V_{out} - V_{in}^{+}}{G_{m1}R_1} \\ &= \pm\frac{V_{OS2}}{G_{m1}R_1} \pm \frac{V_{OS1}}{G_{m1}R_1A_{aux}}\left(\frac{G_{m1}}{G_{m,aux}}\right) \end{aligned} \quad (6)$$

If

$$A_{aux} = \frac{G_{m1}}{G_{m,aux}}, \quad (7)$$

then

$$V_{OS,tot} = \frac{1}{G_{m1}R_1}\left(\pm V_{OS2} \pm V_{OS1}\right). \quad (8)$$

$G_{m1}R_1$ is the first stage of the main amplifier. Its gain is usually on the order of 100. Hence, the offset voltage due to the device mismatch is very small.

As the conventional offset cancellation, the switch S_2 will introduce an injection-induced error. The resulting input-referred offset voltage can be expressed as:

$$V_{OS,inj} = \left(\frac{G_{m,aux}}{G_{m1}}\right)\Delta V. \quad (9)$$

To reduce this error, the value of $G_{m,aux}$ can be chosen to much smaller than that of G_{m1} without increasing $V_{OS,tot}$. Form Equations (8) and (9), a design summary is made: The device sizes of differential pairs of G_{m1} and A_{aux} can be chosen to be larger to obtain a better device match so that V_{OS1} and V_{OS2} will be smaller. Also, a larger value of G_{m1} reduces the offset voltage due to both the device mismatch and clock injection-induced error. The value of A_{aux} is chosen to be on the order of $\left(G_{m1}/G_{m,aux}\right)$.

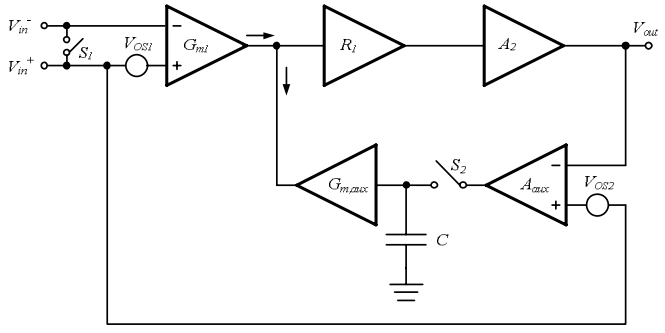


Fig. 2 The proposed offset cancellation architecture.

B. Schematic

The proposed schematic of the two-stage op-amp with cancellation technique is shown in Fig. 3, in which M1~M3, M4~M5, M6~M7, M8, and M9~M13 constitute of G_{m1}, R_1, $A2$, $G_{m,aux}$, and A_{aux}, respectively. G_{m1} and R_1 form the first stage amplifier. Both the G_{m1} and R_1 are differential but $G_{m,aux}$ is single ended. To balance the currents flowing in R_1, the sizes of $G_{m,aux}$ and R_1 are arranged as:

$$\left(\frac{W}{L}\right)_8 + \left(\frac{W}{L}\right)_4 = \left(\frac{W}{L}\right)_5 \quad (10)$$

The aspect ratios of the active load are usually designed to be much smaller than those of the corresponding differential pair. Hence, the transconductance of $G_{m,aux}$ is much smaller than that of G_{m1}. This results in a great reduction of the injection-induced error. Also, the gain of the first stage is easily deigned to be on the order of 100. This means that the offset voltages, which are due to the device mismatches of G_{m1} and A_{aux}, are also greatly reduced.

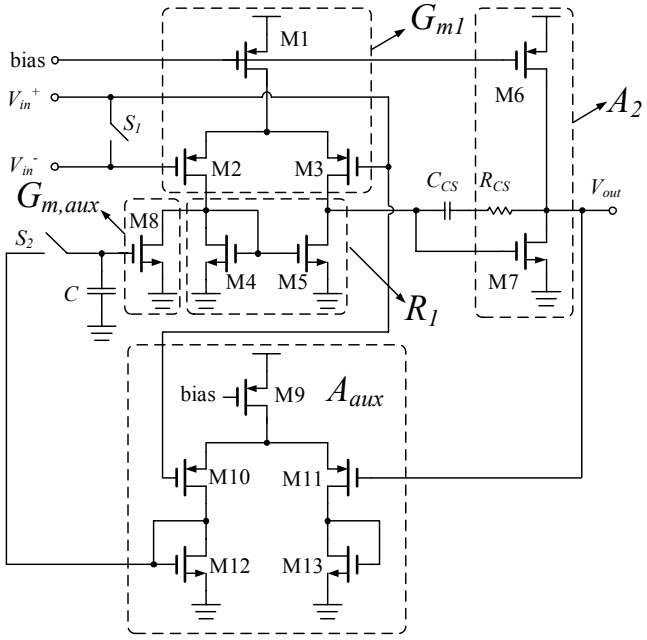


Fig. 3 The proposed schematic of the two-stage op-amp with cancellation technique.

III. EXPERIMENTAL RESULTS

The proposed two-stage op-amp with an offset cancellation technique was fabricated using 0.35-μm CMOS technology. The device sizes used in the amplifier are shown in Table I. The die photograph is shown in Fig. 4. The die area is 125 μm × 140 μm. Fig. 5 shows the measured input and output waveforms for an input of 1 V under a power supply of 3.3 V. The clock frequency is 200 KHz. It can be seen that the offset voltage is 1.898 mV. In this work, the storage capacitor is implemented by a small size of MOS transistor. The offset voltage can be further reduced by increasing the value of the storage capacitor. The performance summary is shown in Table II.

Table I Device sizes ([μm/μm] for transistors).

M1	30/2	M7	120/2	M13	10/2
M2	60/2	M8	40.8/2	*C*	1 pF
M3	240/2	M9	60/2	*Rcs*	4 KΩ
M4	240/2	M10	120/2	*Ccs*	5 pF
M5	5/2	M11	120/2		
M6	10/2	M12	10/2		

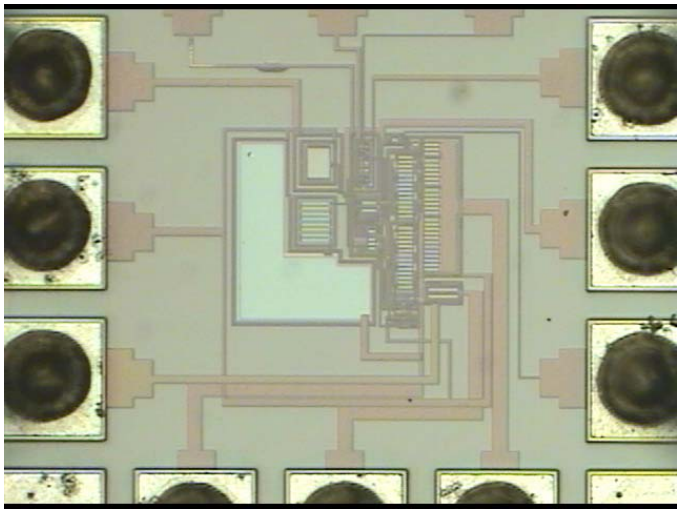


Fig. 4 The die photograph of the proposed two-stage CMOS op-amp with offset cancellation technique.

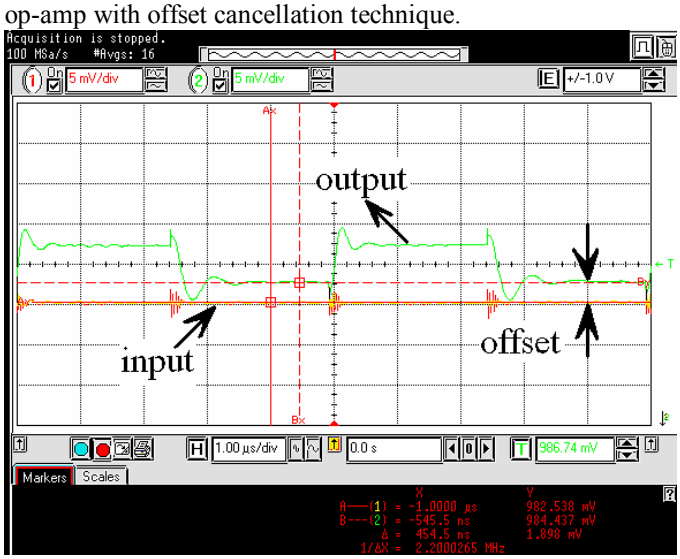


Fig. 5 The measured input and output waveforms for an input of 1 V under a power supply of 3.3 V and 200 KHz clock frequency.

Table II Performance summary.

Process technology	0.35 μm CMOS
VDD	3.3 V
Capacitive load	5 pF
Unity-gain frequency	10 MHz
Phase margin	62°
Power dissipation	0.6 mW
Offset voltage	1.898 mV

IV. Conclusion

An offset cancellation technique, which can be applied in two-stage CMOS op-amps, has been presented. The offset voltages and injection-induced error, which are due to the device mismatches of the circuit and the switch injection respectively, are greatly reduced. An experimental prototype two-stage op-amp implemented in 0.35-μm CMOS technology demonstrates that the offset voltage is only 1.898 mV under a clock frequency of 200 KHz. If the value of the storage capacitor is increased, the offset voltage will be further reduced.

Acknowledgment

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References

[1] Fan You, S.H.K. Embabi and Edgar Sanchez-Sinencio, "Low-Voltage Class AB Output Amplifiers with Quiescent Current Control," IEEE Journal of Solid-State and Circuits. Vol.33, No.6, pp. 915-920, June 1998.

[2] John A. Fisher, "A High-Performance CMOS Power Amplifier" IEEE Journal of Solid-State and Circuits. Vol. SC-20, No.6, pp. 1200-1205, December 1985.

[3] Joongsik Kih, Byungsoo Chang, Deog-Kyoon Jeong, and Wonchan Kim, "Class-AB Large-Swing CMOS Buffer Amplifier with Controlled Bias Current," IEEE Journal of Solid-State and Circuits. Vol. 28, No.12, pp. 1350-1353, December 1993.

[4] Marcel J. M. Pelgrom, Aad C. Duinmaijer, and Anton P. G. Welbers, "Matching Properties of MOS Transistors," IEEE Journal of Solid-State and Circuits. Vol.24, No.5, pp. 1433-1440,October 1989.

[5] Remco J. Wiegerink, Evert Seevinck, and Wim De Jager, "Offset Canceling Circuit," IEEE Journal of Solid-State and Circuits. Vol.24, No.3, pp. 651-658, June 1989.

[6] Marcel J.M. Pelgrom, Hans P. Tuinhout and Maarten Vertregt, "Transistor Matching in Analog CMOS Applications," in 1998 Technical Digest of International Electron Devices Meeting, pp. 915-918.

[7] Behzad Razavi, *Design of Analog CMOS Integrated Circuits*, McGraw-Hill, 2001, pp. 471- 479.